

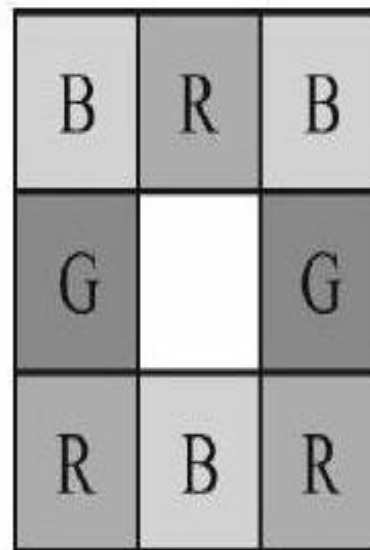
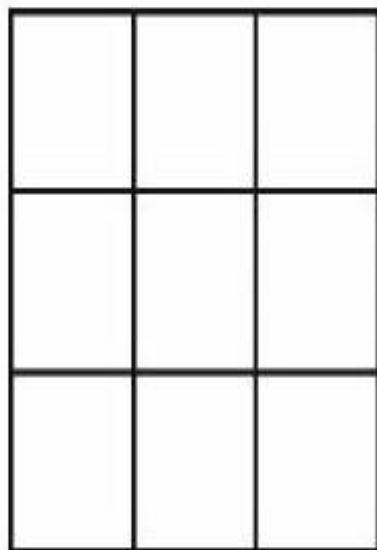
2023 AMC 10A Problem 20 - Solution

Review the full statement and step-by-step solution for 2023 AMC 10A Problem 20. Great practice for AMC 10, AMC 12, AIME, and other math contests

2023 AMC 10A Problem 20

Problem:

Each square in a 3×3 grid of squares is colored red, white, blue, or green so that every 2×2 square contains one square of each color. One such coloring is shown on the right below. How many different colorings are possible?



Answer Choices:

- A. 24
- B. 48
- C. 60

D. 72

E. 96

Solution:

Label the squares as shown.

a	b	c
d	e	f
g	h	i

There are four 2×2 squares that need to satisfy the condition: $abde$, $bcef$, $degh$, and $efhi$.

- Square $abde$: There are $4! = 24$ ways to color these squares using all four colors.
- Square $bcef$: Since squares b and e have already been colored, either c receives the same color as a and f receives the same color as d , or these are switched. This gives 2 choices at this stage. Using the numbers 1, 2, 3, and 4 to represent the colors, the two possibilities are pictured below.

1	2	1
3	4	3

1	2	3
3	4	1

- Remaining squares on the bottom row: Squares d and e have already been colored. If a and c have the same color, then there are two ways to color the three remaining squares:

1	2	1
3	4	3
1	2	1

1	2	1
3	4	3
2	1	2

On the other hand, if squares a and c receive different colors, then there is only one way to color the bottom row:

1	2	3
3	4	1
1	2	3

Thus the number of colorings is $24(2 + 1) = (\text{D}) \boxed{72}$.

The problems and solutions on this page are the property of the MAA's [American Mathematics Competitions](#) ↗

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